Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/27/2025
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	RNA-Seq Guided Mutant Analysis to Discover New Components of Gravity Signaling in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	03/01/2021	**End Date:**	03/01/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	14	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/01/2025 per Y. Zhang/NASA-KSC (Ed., 1/23/25).		
Key Personnel Changes/Previous PI:	None to Report		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0585		
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Performance Goal Text:			

Task Description:	Genes that show contrasting expression across the gravity continuum or between wild-type and gravity perception mutants represent candidates for discovery of novel components of the gravity perception and response pathway in plants. We hypothesize that these genes will reveal novel contributors to gravity perception, signal transduction, and response regulation. We have already completed the necessary RNA-Seq-based contrast analysis and have genes lists in hand to be further studied. The roles and contributions of these genes will be systematically tested in the project proposed here through a series of physiological, cell biological, and molecular screens using knockout mutants. Each mutant will be systematically tested for a gravity-related phenotype using high-resolution image analysis approaches developed in the Principal Investigator's lab. The role of the gene product in controlling differential auxin transport will be assessed by introducing an auxin reporter gene into each mutant background and quantifying expression of the reporter using confocal microscopy. The results of these screens will allow for the mapping of each gene product onto the gravity response pathway, leading to significant insights into plant gravity signaling in support of the development of gravity-altered plants tuned for future spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although gravity is a constant stimulus for plant growth and development, we do not have a full account of how plants detect this stimulus and convert it into molecular, cellular, and physiological information. This research seeks to greatly expand our understanding of how plants sense and respond to this fundamental force. An enhanced understanding of the mechanisms employed by plants to sense and respond to gravity will lead to a more complete and successful set of criteria for engineering crop plants for Earth and future missions off-planet.
Task Progress:	The Starch Statolith theory is the accepted mechanism of gravity sensing in plants and states that dense, starch-filled plastids sediment to the lowest region of the sensing cell. This sedimentation serves as a positional cue and initiates a growth response to reorient the primary root toward the direction of gravity. Despite the primacy of statolith sedimentation, previous research showed that starchless mutants display a persistent, albeit slower, response to gravity. This project identified 124 genes in Arabidopsis that show contrasting expression between wild type and starchless mutants during root gravitropism, representing potential novel contributors to gravity sensing and/or response in roots. The overarching goal of this project was to identify knockout mutants in as many of these genes as possible and test them for differences in gravity response compared to wild type roots. We successfully identified 96 T-DNA knockout mutants and subjected them to a range of experimental tests to determine their root gravity response. Some of the mutants failed to germinate or grow adequately to test, resulting in complete data for 76 of the lines. We carried out 2 main kinds of gravity response experiments. In one type, roots were rotated 90 degrees from vertical and allowed to respond back toward vertical while images were captured every 10 min for 220 min. In this kind of experiment, called a free response, 26 of the mutants showed a response that differed significantly from the wild type, with most showing a reduced response. In the other type of experiment, roots were initially rotated 90 degrees from vertical as in the free response type, but were then continuously rotated at a rate of 15 deg/h. This experiment was done to test the root's ability to respond to continuous stimulation. In this continuous rotation experiment, 20 mutants showed a response that differed significantly from the wild type. Interestingly, only 8 of the 26 mutants showing a significant difference in the free response experiment also showed a significant response in the continuous rotation experiments, indicating the effectiveness of the two assays at identifying different aspects of the gravity response pathway. Mutants that showed a significant gravity response difference were crossed into the starchless mutant background to begin the process of making double mutants for analysis. It is hypothesized that if a gene product contributes to the starchless plants' ability to sense or respond to gravity, this ability will be more severely affected in the starchless root background. Of the 38 total mutants, successful F1 seeds have been harvested for 18 thus far and work is continuing on this objective. Of the F1 lines confirmed, 6 have been carried out to the F3 generation for scoring and identification of double homozygous alleles. These lines will be subjected to the same free response and continuous rotation experiments after they are obtained. In addition to creating double mutants with significant mutants, they were also crossed into the R2D2 semi-quantitative auxin reporter line. One of the regulators of growth and gravitropism, the cellular levels of the plant hormone auxin can be inferred using this fluorescent reporter. The fluorescent reporter protein in this system is targeted for degradation when auxin enters a cell, thus allowing the tracking of its disappearance as an indicator of auxin transport. The reporter also includes a second fluorescent protein that is not able to be targeted for degradation, thus allowing the ratiometric measurement of the two proteins as an internal control. Thus far, 12 F1 lines have been identified and 4 F3 lines have been used for pilot research. This molecular tool will allow for the determination of auxin transport in the mutants and the mapping of the gene product to a point in the gravity response pathway.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Wolverton C. "Transcriptomic overview of plant responses across a continuum of gravity treatments." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Articles in Peer-reviewed Journals	Roychoudhry S, Sageman-Furnas K, Taylor HJ, Showpnil I, Wolverton C, Friml J, Del Bianco M, Kepinski S. "Angle dependence as a unifying feature of root graviresponse modules." Proceedings of the National Academy of Sciences. 2025 Nov 18;122(46):e2506400122. https://doi.org/10.1073/pnas.2506400122 , Nov-2025
Articles in Peer-reviewed Journals	Roychoudhry S, Sageman-Furnas K, Taylor HJ, Showpnil I, Wolverton C, Friml J, Del Bianco M, Kepinski S. "Angle dependence as a unifying feature of root graviresponse modules." Proc Natl Acad Sci USA. 2025 Nov 18;122(46):e2506400122. https://doi.org/10.1073/pnas.2506400122 ; PMID: 41218119 ; PMCID: PMC12646229 , Nov-2025
Articles in Peer-reviewed Journals	Canaday ELJ, Meyers A, Scinto-Madonich NJ, Wyatt SE, Wolverton C. "Physiological and transcriptomic assessment of Arabidopsis identifies two distinct gravity signaling systems." Plant Physiol. 2026 Feb 27;kiag095. Online ahead of print. https://doi.org/10.1093/plphys/kiag095 ; PMID: 41761687 , Feb-2026